REVIEW
WORKFORCE ANALYTICS: EMPLOYEE ATTRITION ANALYSIS

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Student: M Nithin
Domain: Human Resource Analytics

Introduction

Employee attrition is one of the key challenges faced by organizations as it impacts productivity, operational costs, and workforce stability. Understanding the factors that influence employee turnover is essential for improving employee retention and organizational performance. This project focuses on preparing and cleaning employee-related data to ensure its quality and reliability for future attrition analysis. The study aims to establish a strong data foundation that supports meaningful insights and informed business decisions.

Problem Statement

Organizations often struggle to identify the factors contributing to employee attrition due to the presence of poor-quality data, including missing values, duplicate records, inconsistencies, and outliers. These issues can affect the accuracy of analysis and decision-making. This project aims to assess, clean, and standardize employee data to create a reliable dataset that can be used for future attrition analysis and workforce retention strategies.

Project Objectives

- Understand the employee dataset.
- Assess data quality issues.
- Perform data cleaning and preprocessing.
- Import cleaned data into PostgreSQL.
- Perform SQL-based attrition analysis.
- Generate business insights.
- Prepare data for Power BI dashboard development.

Dataset Overview

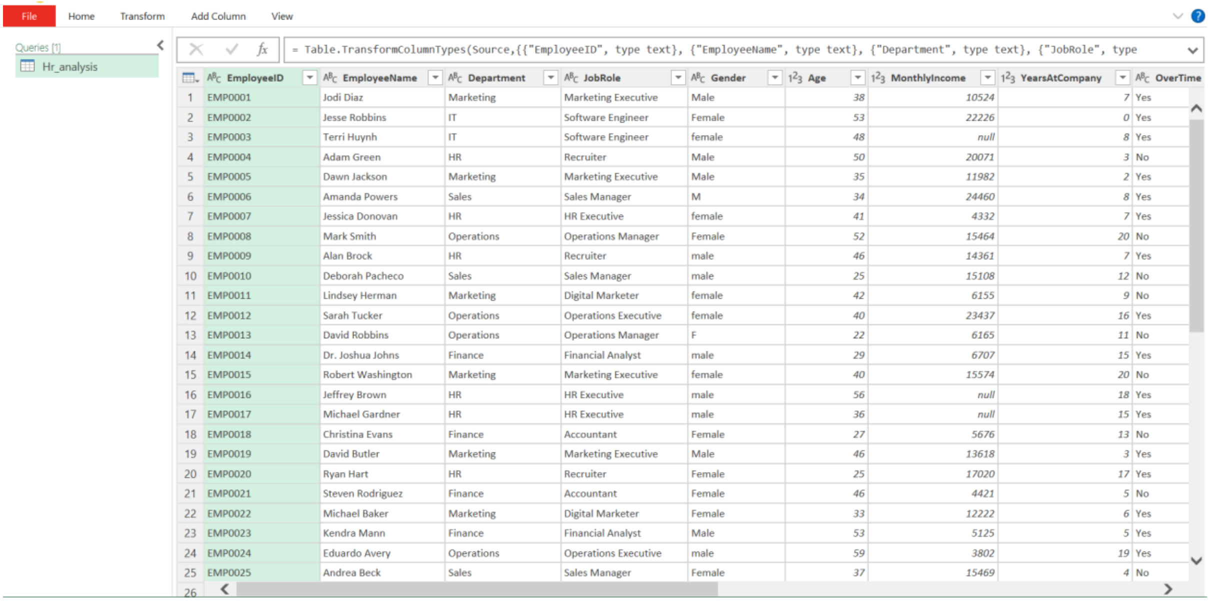
- Dataset Name: Employee Attrition Dataset
- Number of Records: 1225
- Number of Attributes: 15
- Target Variable: Attrition
- Granularity: One row represents one employee.

Data Dictionary

Attribute	Description
EmployeeID	Employee Identifier
EmployeeName	Employee Name
Department	Employee Department
JobRole	Employee Role
Gender	Employee Gender
Age	Employee Age
MonthlyIncome	Employee Monthly Income
YearsAtCompany	Employee Experience
OverTime	Overtime Status
JobSatisfaction	Satisfaction Rating
WorkLifeBalance	Work-Life Balance Rating
EnvironmentSatisfaction	Environment Satisfaction Rating
DistanceFromHome	Distance from Home
MaritalStatus	Marital Status
Attrition	Employee Attrition Status

DATA CLEANING AND PREPROCESSING

Dataset Loading


File Home Transform Add Column View

Queries [1]

Hr_analysis

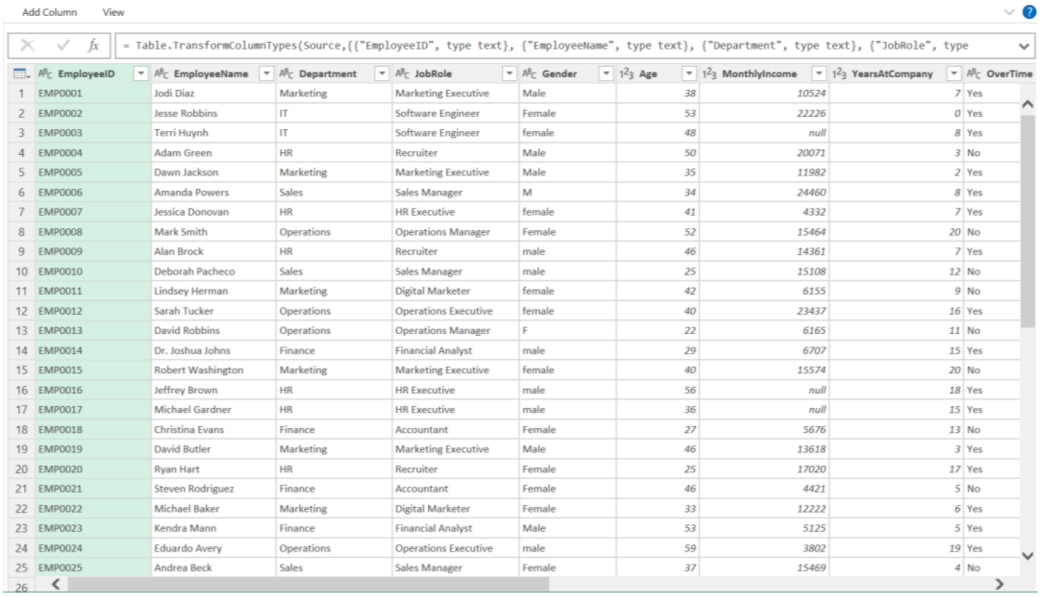
= Table.TransformColumnTypes(Source,{{"EmployeeID", type text}, {"EmployeeName", type text}, {"Department", type text}, {"JobRole", type

	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38	10524	7	Yes
2	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53	22226	0	Yes
3	EMP0003	Terri Huynh	IT	Software Engineer	female	48	null	8	Yes
4	EMP0004	Adam Green	HR	Recruiter	Male	50	20071	3	No
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35	11982	2	Yes
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	8	Yes
7	EMP0007	Jessica Donovan	HR	HR Executive	female	41	4332	7	Yes
8	EMP0008	Mark Smith	Operations	Operations Manager	Female	52	15464	20	No
9	EMP0009	Alan Brock	HR	Recruiter	male	46	14361	7	Yes
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25	15108	12	No
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42	6155	9	No
12	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40	23437	16	Yes
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	6165	11	No
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29	6707	15	Yes
15	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40	15574	20	No
16	EMP0016	Jeffrey Brown	HR	HR Executive	male	56	null	18	Yes
17	EMP0017	Michael Gardner	HR	HR Executive	male	36	null	15	Yes
18	EMP0018	Christina Evans	Finance	Accountant	Female	27	5676	13	No
19	EMP0019	David Butler	Marketing	Marketing Executive	Male	46	13618	3	Yes
20	EMP0020	Ryan Hart	HR	Recruiter	Female	25	17020	17	Yes
21	EMP0021	Steven Rodriguez	Finance	Accountant	Female	46	4421	5	No
22	EMP0022	Michael Baker	Marketing	Digital Marketer	Female	33	12222	6	Yes
23	EMP0023	Kendra Mann	Finance	Financial Analyst	Male	53	5125	5	Yes
24	EMP0024	Eduardo Avery	Operations	Operations Executive	male	59	3802	19	Yes
25	EMP0025	Andrea Beck	Sales	Sales Manager	Female	37	15469	4	No
26									

Description

- Employee attrition dataset was loaded into Excel and Power Query.
- Dataset structure and attributes were reviewed.
- Initial data assessment was performed.

Data Profiling



	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38	10524	7	Yes
2	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53	22226	0	Yes
3	EMP0003	Terri Huynh	IT	Software Engineer	female	48	null	8	Yes
4	EMP0004	Adam Green	HR	Recruiter	Male	50	20071	3	No
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35	11982	2	Yes
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	8	Yes
7	EMP0007	Jessica Donovan	HR	HR Executive	female	41	4332	7	Yes
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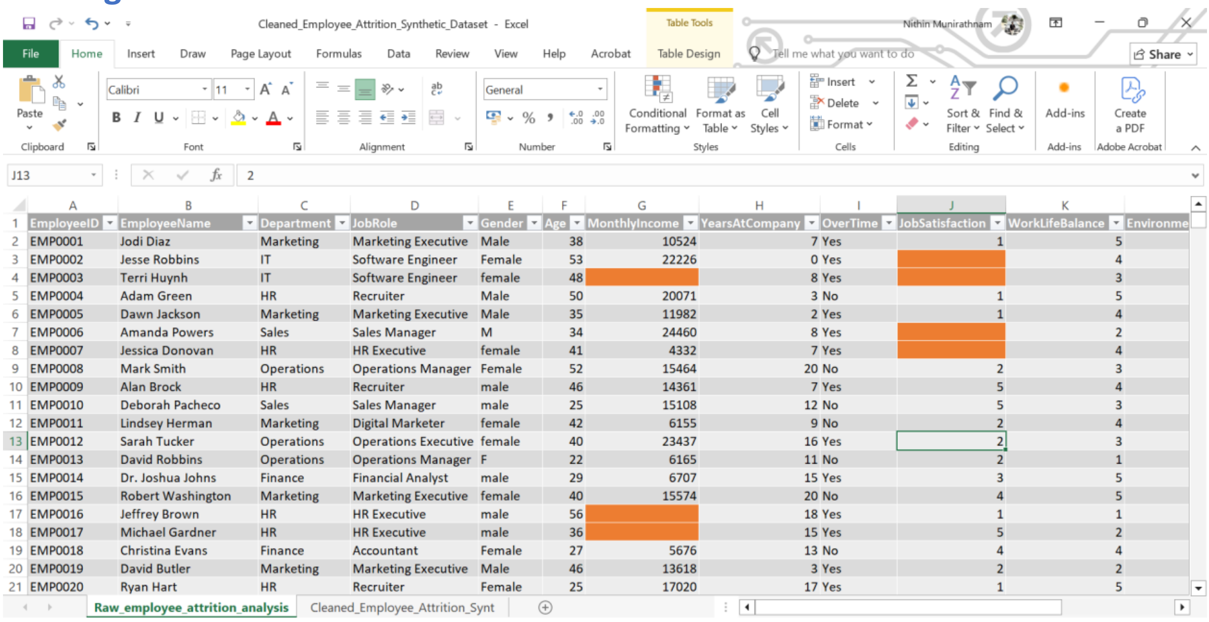
9	EMP0009	Alan Brock	HR	Recruiter	male	46	14361	7	Yes
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25	15108	12	No
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42	6155	9	No
12	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40	23437	16	Yes
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	6165	11	No
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29	6707	15	Yes
15	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40	15574	20	No
16	EMP0016	Jeffrey Brown	HR	HR Executive	male	56	null	18	Yes
17	EMP0017	Michael Gardner	HR	HR Executive	male	36	null	15	Yes
18	EMP0018	Christina Evans	Finance	Accountant	Female	27	5676	13	No
19	EMP0019	David Butler	Marketing	Marketing Executive	Male	46	13618	3	Yes
20	EMP0020	Ryan Hart	HR	Recruiter	Female	25	17020	17	Yes
21	EMP0021	Steven Rodriguez	Finance	Accountant	Female	46	4421	5	No
22	EMP0022	Michael Baker	Marketing	Digital Marketer	Female	33	12222	6	Yes
23	EMP0023	Kendra Mann	Finance	Financial Analyst	Male	53	5125	5	Yes
24	EMP0024	Eduardo Avery	Operations	Operations Executive	male	59	3802	19	Yes
25	EMP0025	Andrea Beck	Sales	Sales Manager	Female	37	15469	4	No

Description

- Dataset was profiled to understand attribute distributions.
- Data types and quality issues were examined.
- Potential cleaning requirements were identified.

Missing Value Identification

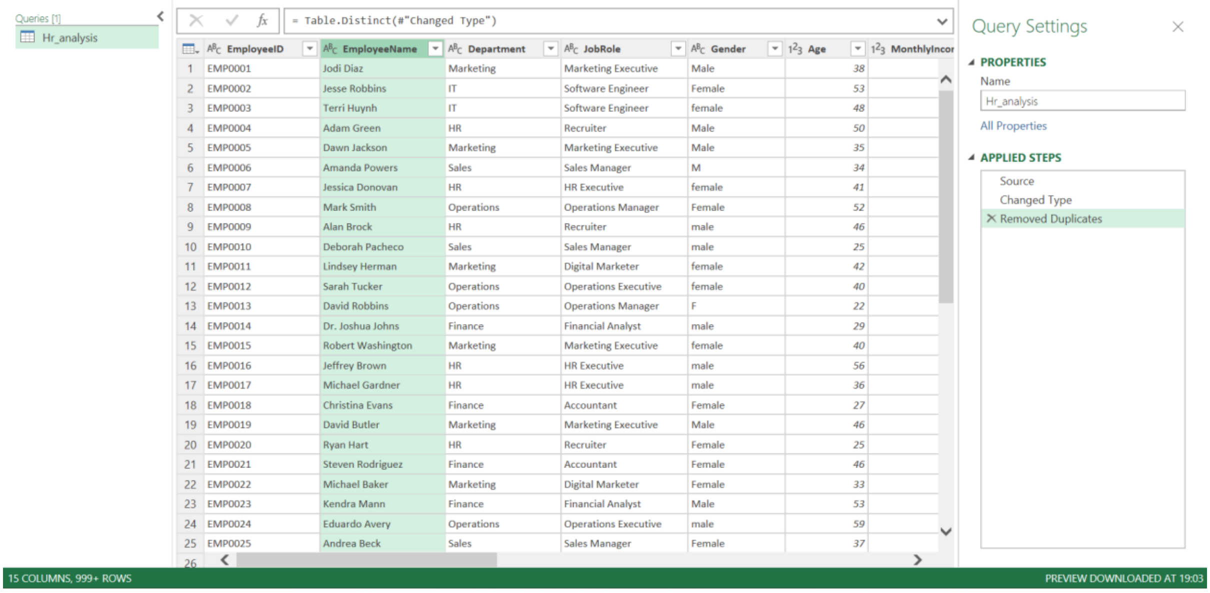


	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime	JobSatisfaction	WorkLifeBalance	Environme
2	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38	10524	7	Yes	1	5	
3	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53	22226	0	Yes		4	
4	EMP0003	Terri Huynh	IT	Software Engineer	female	48		8	Yes		3	
5	EMP0004	Adam Green	HR	Recruiter	Male	50	20071	3	No	1	5	
6	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35	11982	2	Yes	1	4	
7	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	8	Yes		2	
8	EMP0007	Jessica Donovan	HR	HR Executive	female	41	4332	7	Yes		4	
9	EMP0008	Mark Smith	Operations	Operations Manager	Female	52	15464	20	No	2	3	
10	EMP0009	Alan Brock	HR	Recruiter	male	46	14361	7	Yes	5	4	
11	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25	15108	12	No	5	3	
12	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42	6155	9	No	2	4	
13	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40	23437	16	Yes	2	3	
14	EMP0013	David Robbins	Operations	Operations Manager	F	22	6165	11	No	2	1	
15	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29	6707	15	Yes	3	5	
16	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40	15574	20	No	4	5	
17	EMP0016	Jeffrey Brown	HR	HR Executive	male	56		18	Yes	1	1	
18	EMP0017	Michael Gardner	HR	HR Executive	male	36		15	Yes	5	2	
19	EMP0018	Christina Evans	Finance	Accountant	Female	27	5676	13	No	4	4	
20	EMP0019	David Butler	Marketing	Marketing Executive	Male	46	13618	3	Yes	2	2	
21	EMP0020	Ryan Hart	HR	Recruiter	Female	25	17020	17	Yes	1	5	

Description

- Missing values were identified in multiple attributes.
- Missing data can impact analytical accuracy.
- Missing values were documented for treatment.

Removed duplicates:

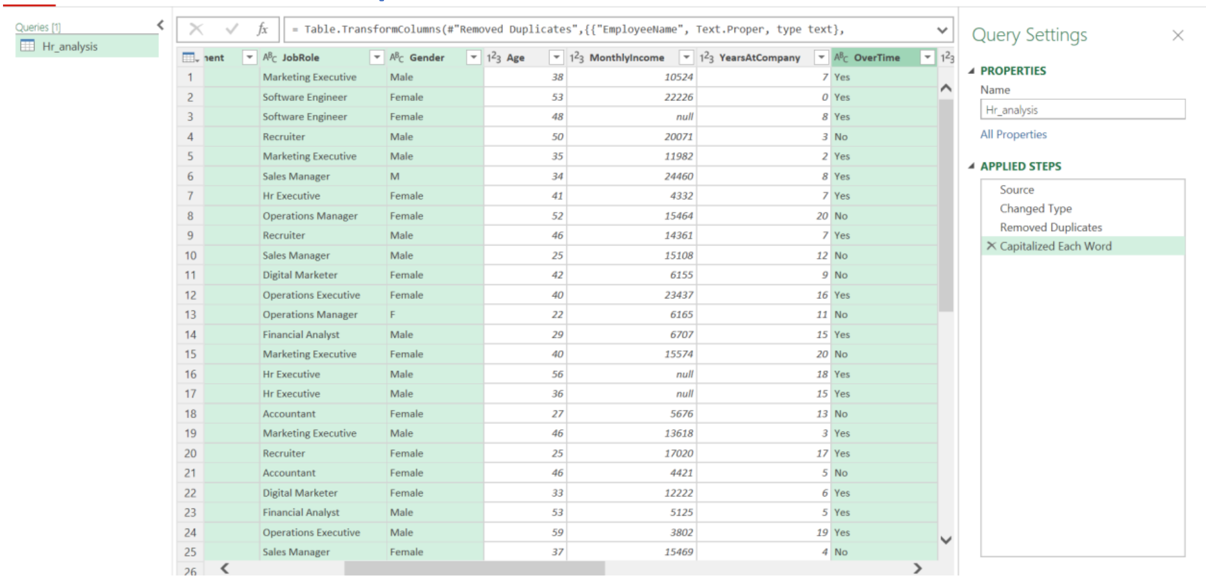


	EmployeeID	EmployeeName	Department	JobRole	Gender	Age
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38
2	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53
3	EMP0003	Terri Huynh	IT	Software Engineer	female	48
4	EMP0004	Adam Green	HR	Recruiter	Male	50
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34
7	EMP0007	Jessica Donovan	HR	HR Executive	female	41
8	EMP0008	Mark Smith	Operations	Operations Manager	Female	52
9	EMP0009	Alan Brock	HR	Recruiter	male	46
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42
12	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40
13	EMP0013	David Robbins	Operations	Operations Manager	F	22
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29
15	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40
16	EMP0016	Jeffrey Brown	HR	HR Executive	male	56
17	EMP0017	Michael Gardner	HR	HR Executive	male	36
18	EMP0018	Christina Evans	Finance	Accountant	Female	27
19	EMP0019	David Butler	Marketing	Marketing Executive	Male	46
20	EMP0020	Ryan Hart	HR	Recruiter	Female	25
21	EMP0021	Steven Rodriguez	Finance	Accountant	Female	46
22	EMP0022	Michael Baker	Marketing	Digital Marketer	Female	33
23	EMP0023	Kendra Mann	Finance	Financial Analyst	Male	53
24	EMP0024	Eduardo Avery	Operations	Operations Executive	male	59
25	EMP0025	Andrea Beck	Sales	Sales Manager	Female	37

Description

- The dataset was examined for duplicate employee records.
- Duplicate entries can lead to inaccurate employee counts and misleading analytical results.
- All identified duplicate records were removed to maintain data integrity and consistency.

Data Standardization Capitalization:

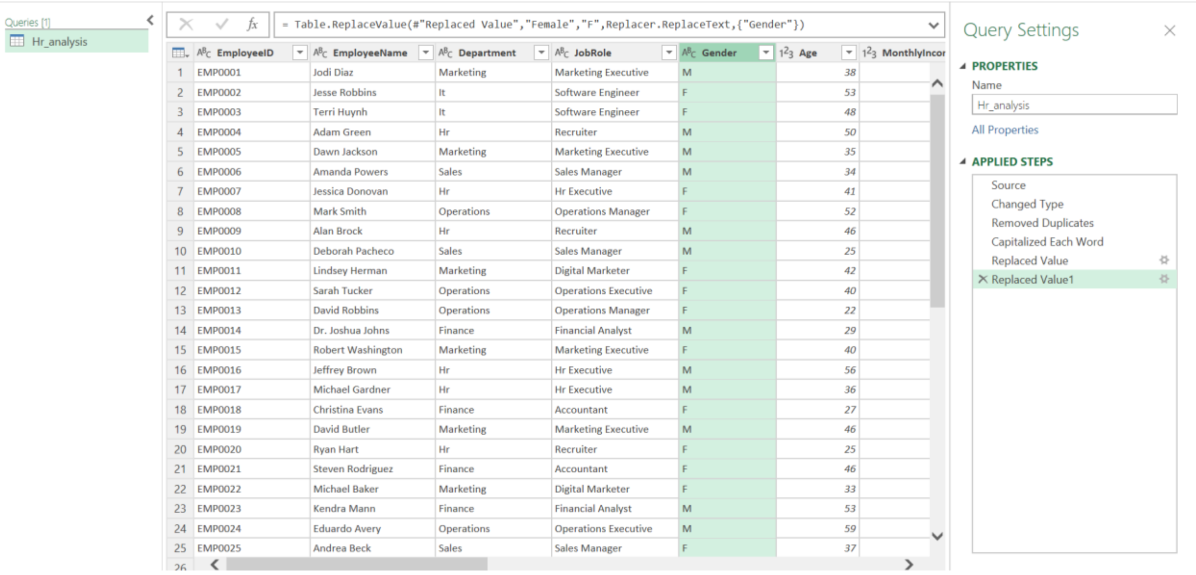


	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime
1	Marketing Executive	Male	38	10524	7	Yes
2	Software Engineer	Female	53	22226	0	Yes
3	Software Engineer	Female	48	null	8	Yes
4	Recruiter	Male	50	20071	3	No
5	Marketing Executive	Male	35	11982	2	Yes
6	Sales Manager	M	34	24460	8	Yes
7	Hr Executive	Female	41	4332	7	Yes
8	Operations Manager	Female	52	15464	20	No
9	Recruiter	Male	46	14361	7	Yes
10	Sales Manager	Male	25	15108	12	No
11	Digital Marketer	Female	42	6155	9	No
12	Operations Executive	Female	40	23437	16	Yes
13	Operations Manager	F	22	6165	11	No
14	Financial Analyst	Male	29	6707	15	Yes
15	Marketing Executive	Female	40	15574	20	No
16	Hr Executive	Male	56	null	18	Yes
17	Hr Executive	Male	36	null	15	Yes
18	Accountant	Female	27	5676	13	No
19	Marketing Executive	Male	46	13618	3	Yes
20	Recruiter	Female	25	17020	17	Yes
21	Accountant	Female	46	4421	5	No
22	Digital Marketer	Female	33	12222	6	Yes
23	Financial Analyst	Male	53	5125	5	Yes
24	Operations Executive	Male	59	3802	19	Yes
25	Sales Manager	Female	37	15469	4	No

Description

- Text values were standardized using capitalization rules.
- Inconsistent text formats were corrected to improve data consistency.
- Standardized values help ensure accurate grouping and reporting during analysis.

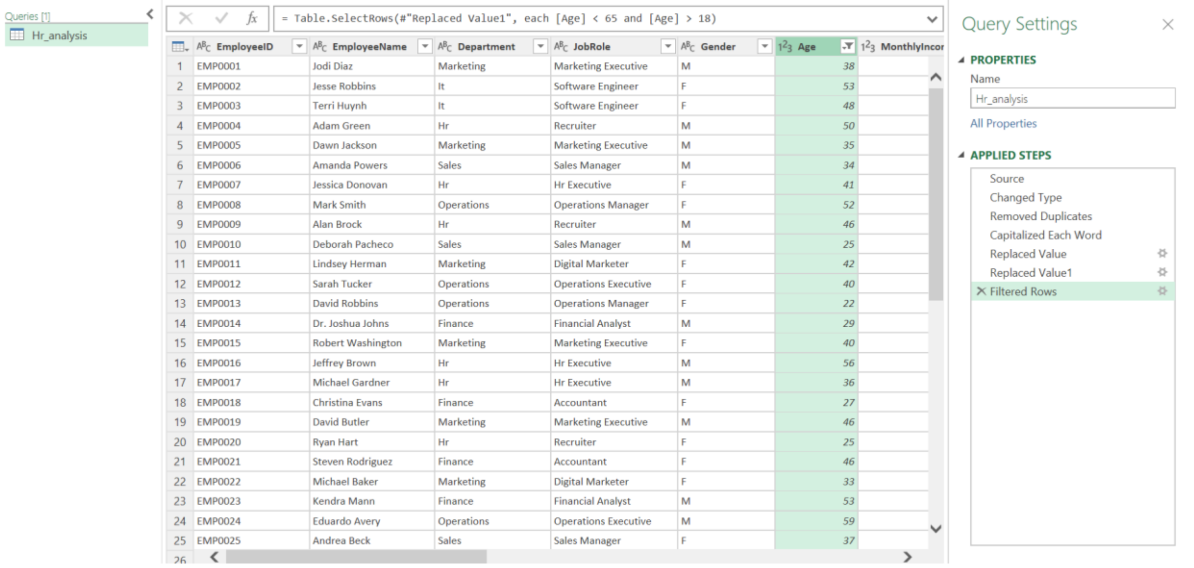
Gender column



Description

- Inconsistent gender representations were identified in the dataset.
- Values were standardized into a consistent format to avoid duplicate categories.
- This transformation improves the accuracy of gender-based analysis and reporting.

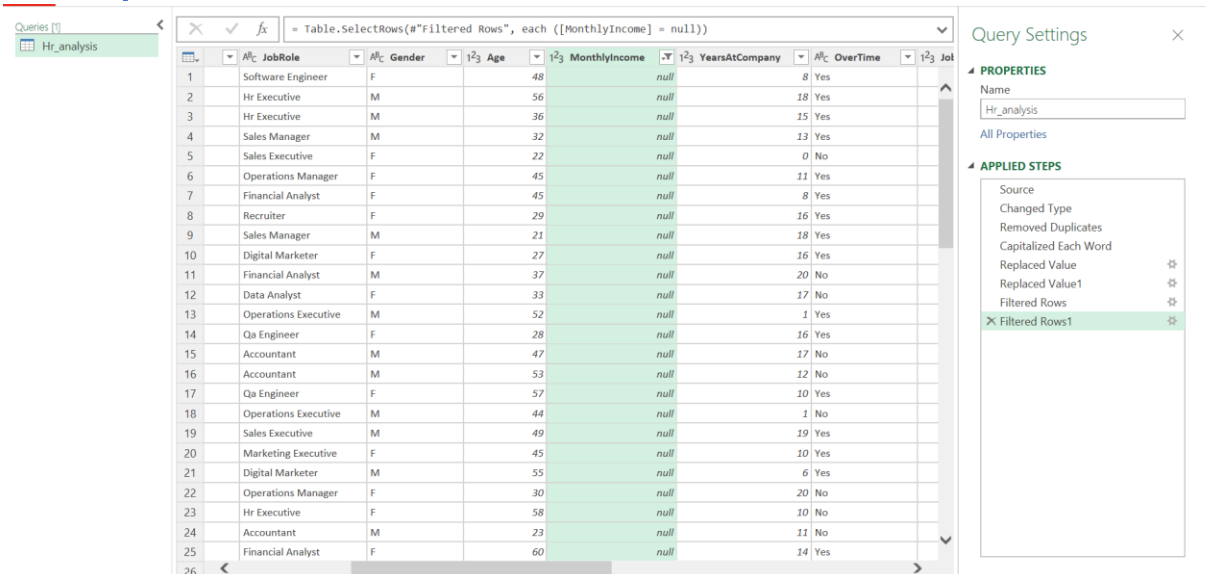
Age Column



Description

- Invalid age values were identified during data quality assessment.
- Employee records with age values outside the acceptable business range were treated as outliers.
- Outlier records were removed using filtering techniques to improve data reliability.

Monthly income:

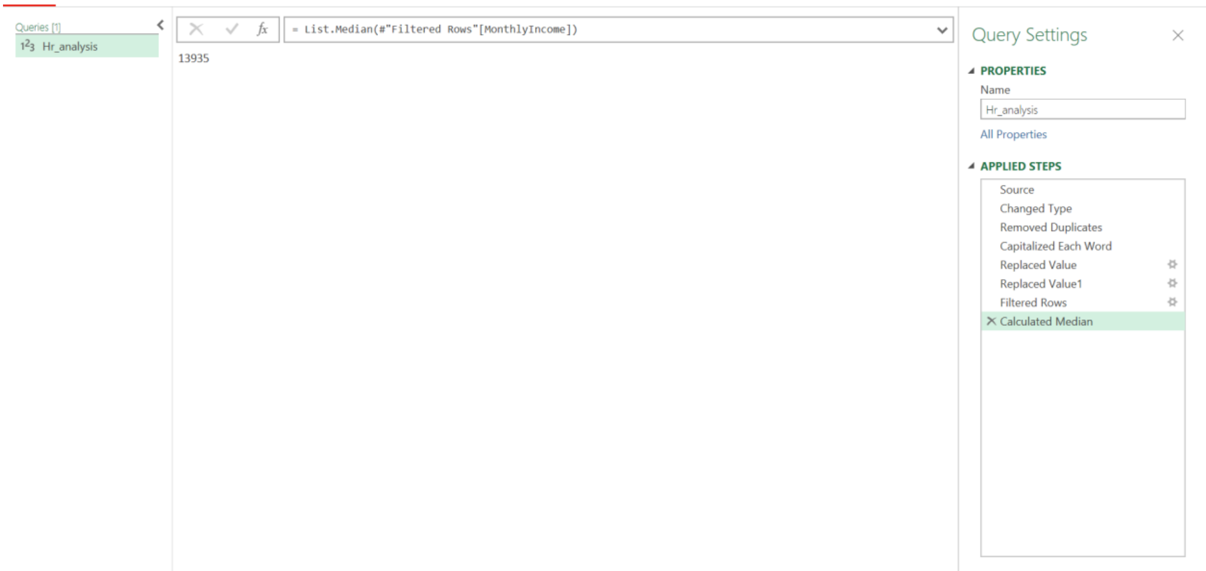


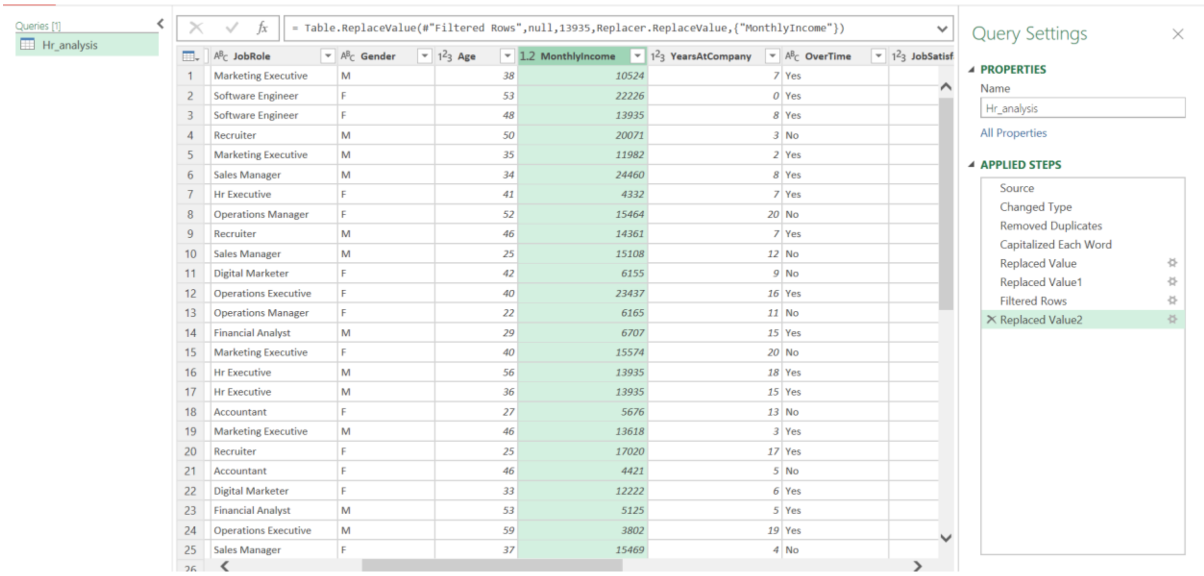
Description

- Missing values were identified in the Monthly Income attribute.
- Null values can negatively affect statistical analysis and reporting.
- Appropriate treatment was required to maintain dataset completeness.

Removed null values by median imputation:

Calculated median:



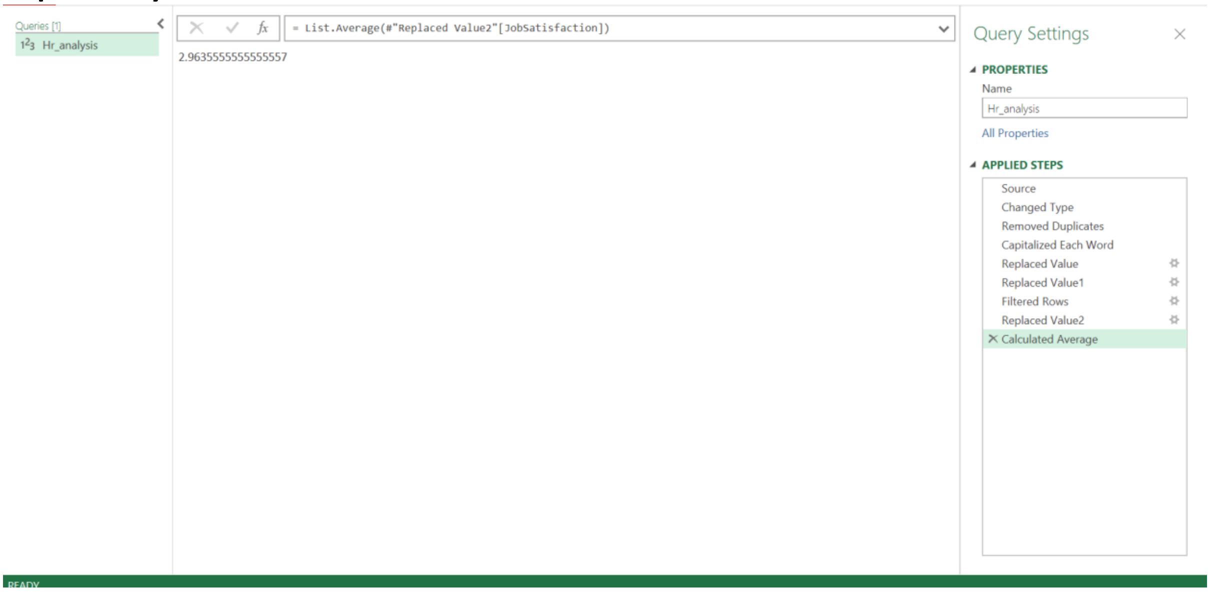


Description

- The median value of Monthly Income was calculated.
- Missing income values were replaced using median imputation.
- Median was selected because it is less affected by extreme income values and provides a more representative measure of central tendency.

Job satisfaction:

Replaced by mean and rounded of to 3

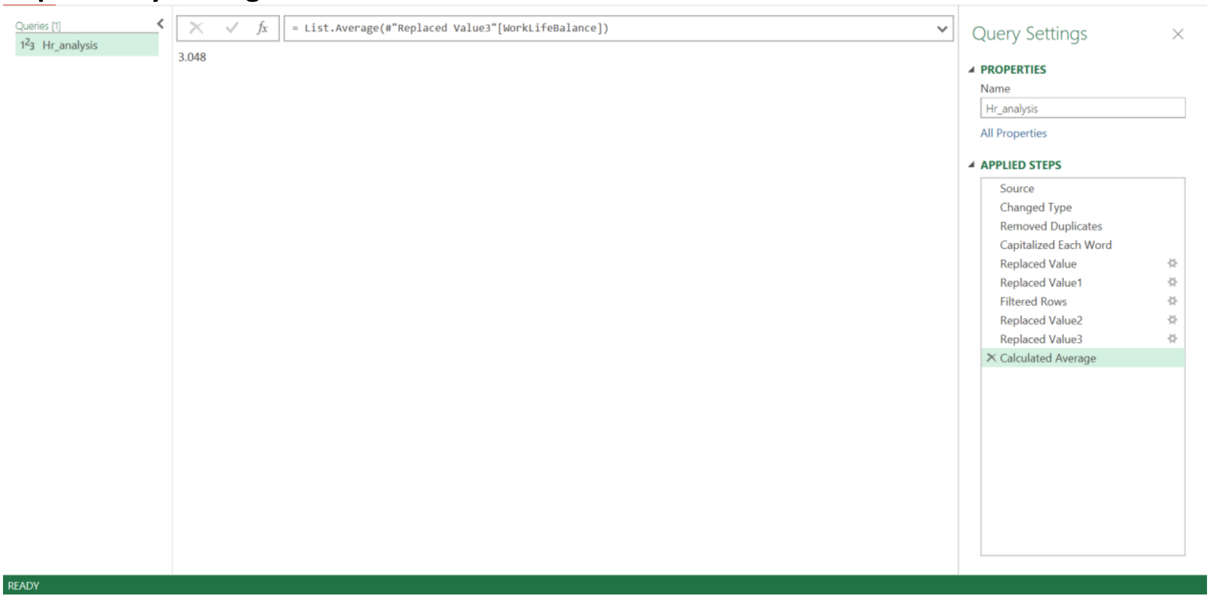


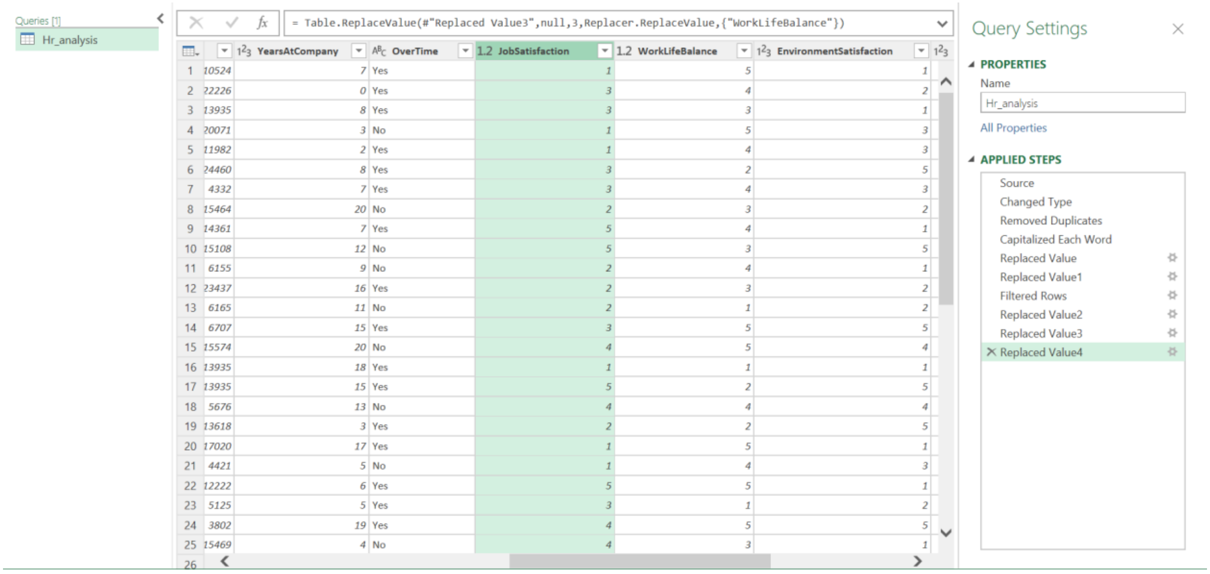
Description

- Missing values were identified in the JobSatisfaction attribute.
- The average value was calculated and rounded to the nearest whole number (3).
- Missing values were replaced with the rounded average value to maintain data consistency.

Work life balance:

Replaced by average and rounded of to 3

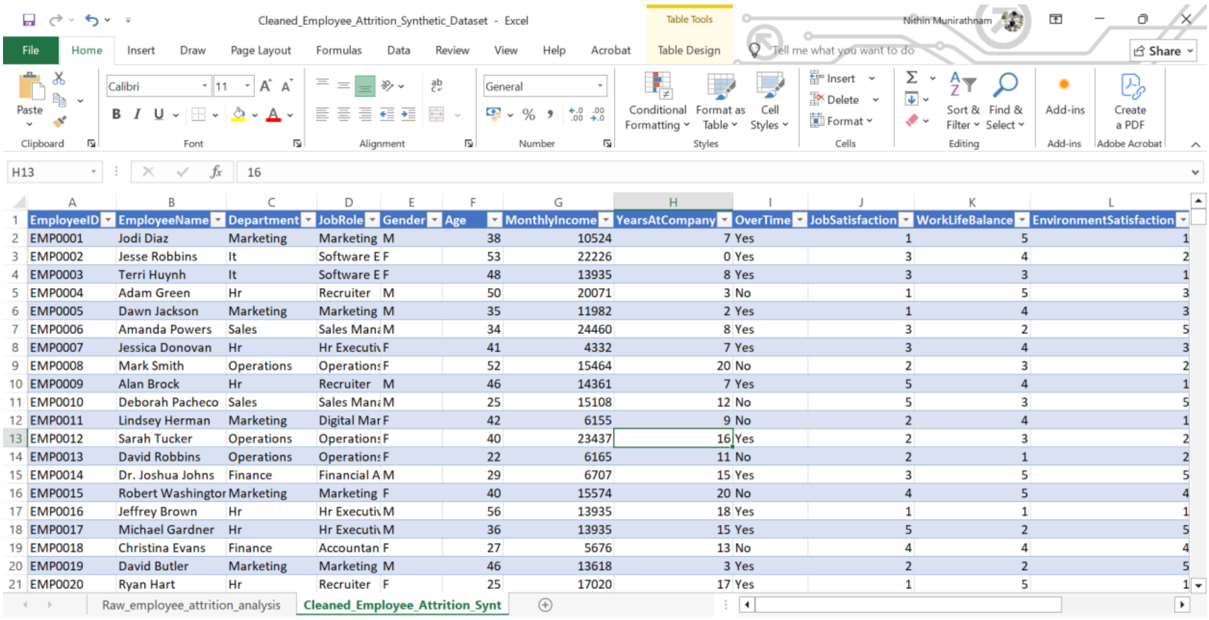




Description

- Missing values were identified in the WorkLifeBalance attribute.
- The average value was calculated and rounded to the nearest whole number (3).
- Missing values were replaced using the rounded average value to preserve the overall distribution of the data.

Final Data Validation



EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime	JobSatisfaction	WorkLifeBalance	EnvironmentSatisfaction
EMP0001	Jodi Diaz	Marketing	Marketing	M	38	10524	7	Yes	1	5	1
EMP0002	Jesse Robbins	It	Software E	F	53	22226	0	Yes	3	4	2
EMP0003	Terri Huynh	It	Software E	F	48	13935	8	Yes	3	3	1
EMP0004	Adam Green	Hr	Recruiter	M	50	20071	3	No	1	5	3
EMP0005	Dawn Jackson	Marketing	Marketing	M	35	11982	2	Yes	1	4	3
EMP0006	Amanda Powers	Sales	Sales Man	M	34	24460	8	Yes	3	2	5
EMP0007	Jessica Donovan	Hr	Hr Executi	F	41	4332	7	Yes	3	4	3
EMP0008	Mark Smith	Operations	Operation	F	52	15464	20	No	2	3	2
EMP0009	Alan Brock	Hr	Recruiter	M	46	14361	7	Yes	5	4	1
EMP0010	Deborah Pacheco	Sales	Sales Man	M	25	15108	12	No	5	3	5
EMP0011	Lindsey Herman	Marketing	Digital Mar	F	42	6155	9	No	2	4	1
EMP0012	Sarah Tucker	Operations	Operation	F	40	23437	16	Yes	2	3	2
EMP0013	David Robbins	Operations	Operation	F	22	6165	11	No	2	1	2
EMP0014	Dr. Joshua Johns	Finance	Financial A	M	29	6707	15	Yes	3	5	5
EMP0015	Robert Washingtor	Marketing	Marketing	F	40	15574	20	No	4	5	4
EMP0016	Jeffrey Brown	Hr	Hr Executi	M	56	13935	18	Yes	1	1	1
EMP0017	Michael Gardner	Hr	Hr Executi	M	36	13935	15	Yes	5	2	5
EMP0018	Christina Evans	Finance	Accountan	F	27	5676	13	No	4	4	4
EMP0019	David Butler	Marketing	Marketing	M	46	13618	3	Yes	2	2	5
EMP0020	Ryan Hart	Hr	Recruiter	F	25	17020	17	Yes	1	5	1

Description

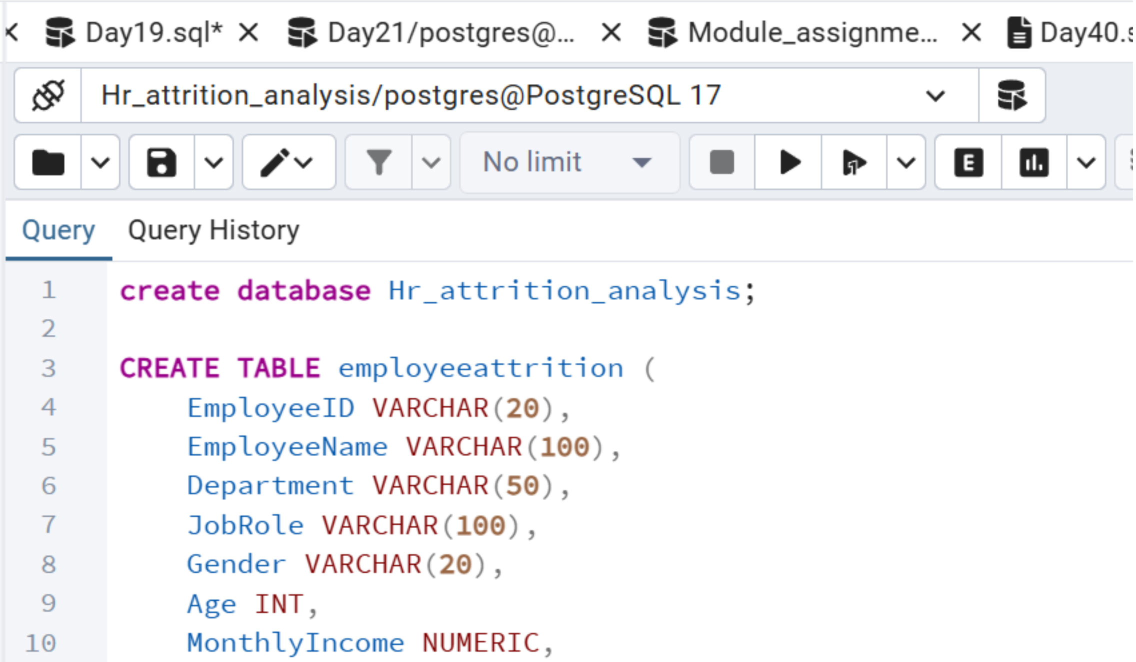
- The cleaned dataset was reviewed to verify all transformations.
- Missing values, duplicate records, inconsistencies, and outliers were successfully addressed.
- The dataset was validated and prepared for SQL-based analysis.

Export of Cleaned Dataset

- Cleaned dataset was exported as CSV format.
- CSV file was prepared for PostgreSQL import.
- Data was validated before import.

SQL ANALYSIS

PostgreSQL Database Creation



Description

- HR_Attrition_Analysis database was created.
- Database environment was configured.
- Prepared for storing employee data.

16. Table Creation

Hr_attrition_analysis/postgres@PostgreSQL 17

No limit

Query Query History

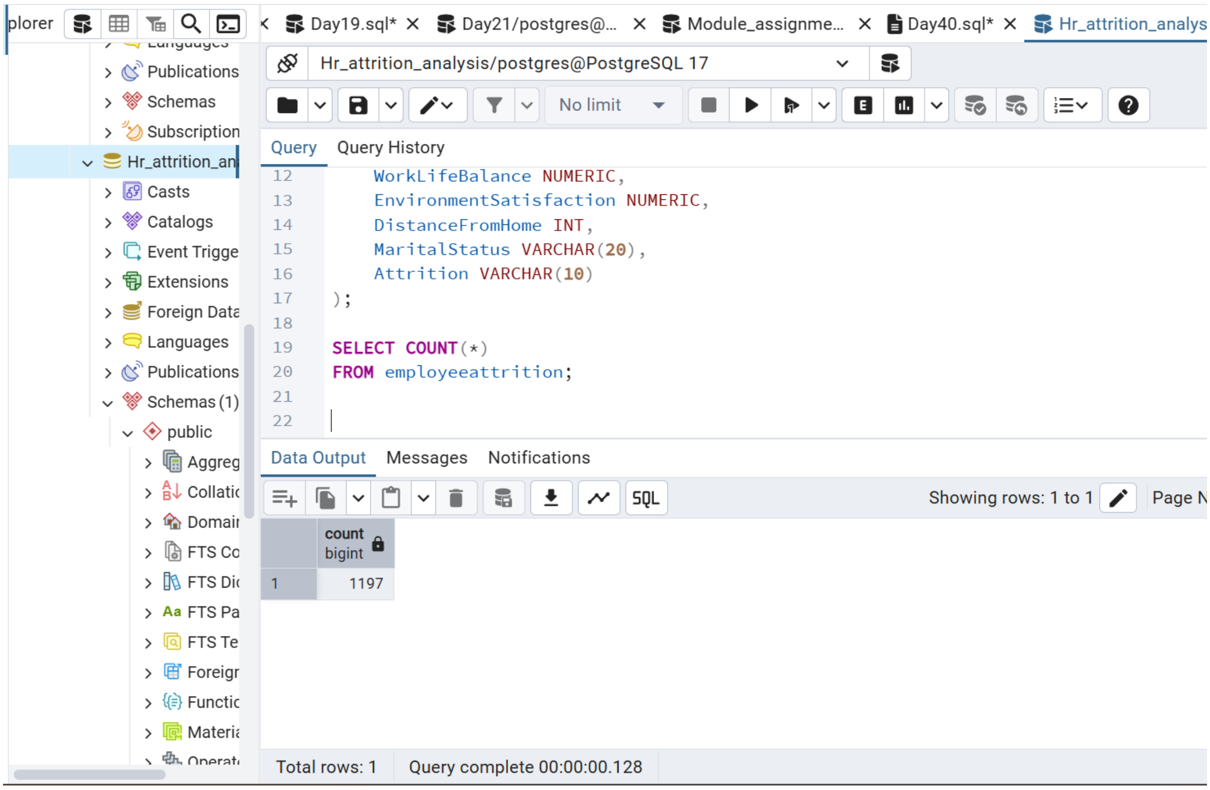
```sql
create database Hr_attrition_analysis;

CREATE TABLE employeeattrition (
    EmployeeID VARCHAR(20),
    EmployeeName VARCHAR(100),
    Department VARCHAR(50),
    JobRole VARCHAR(100),
    Gender VARCHAR(20),
    Age INT,
    MonthlyIncome NUMERIC,
    YearsAtCompany INT,
    OverTime VARCHAR(10),
    JobSatisfaction NUMERIC,
    WorkLifeBalance NUMERIC,
    EnvironmentSatisfaction NUMERIC,
    DistanceFromHome INT,
    MaritalStatus VARCHAR(20),
    Attrition VARCHAR(10)
);
```

Description

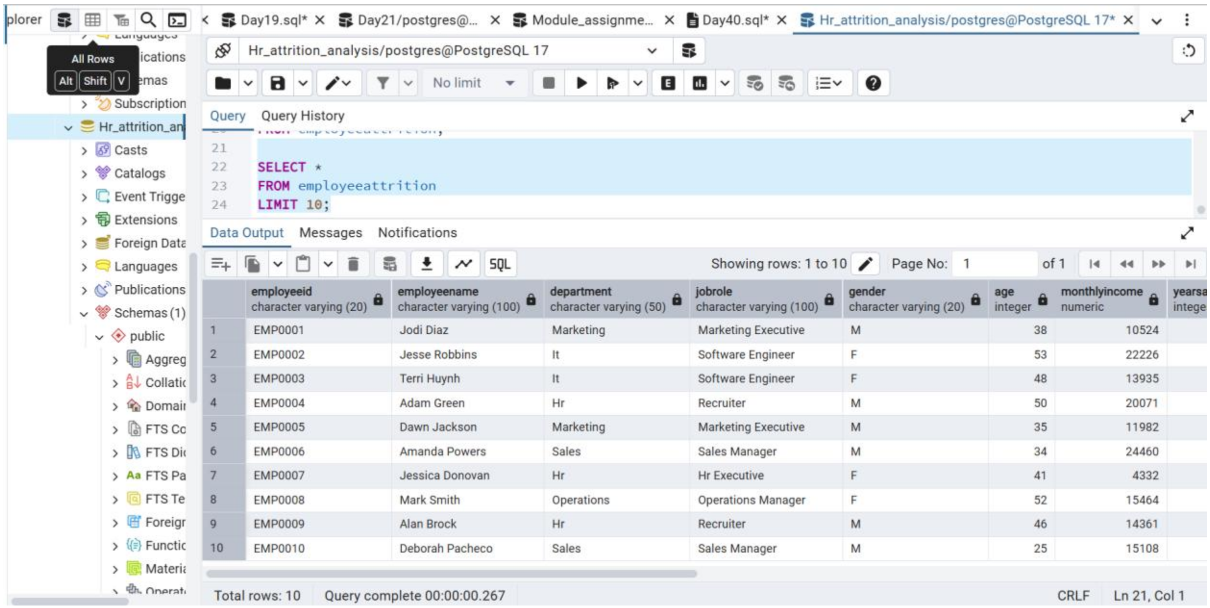
- Employee Attrition table was created.
- Appropriate data types were assigned.
- Table structure was validated.

Data validation

Total records

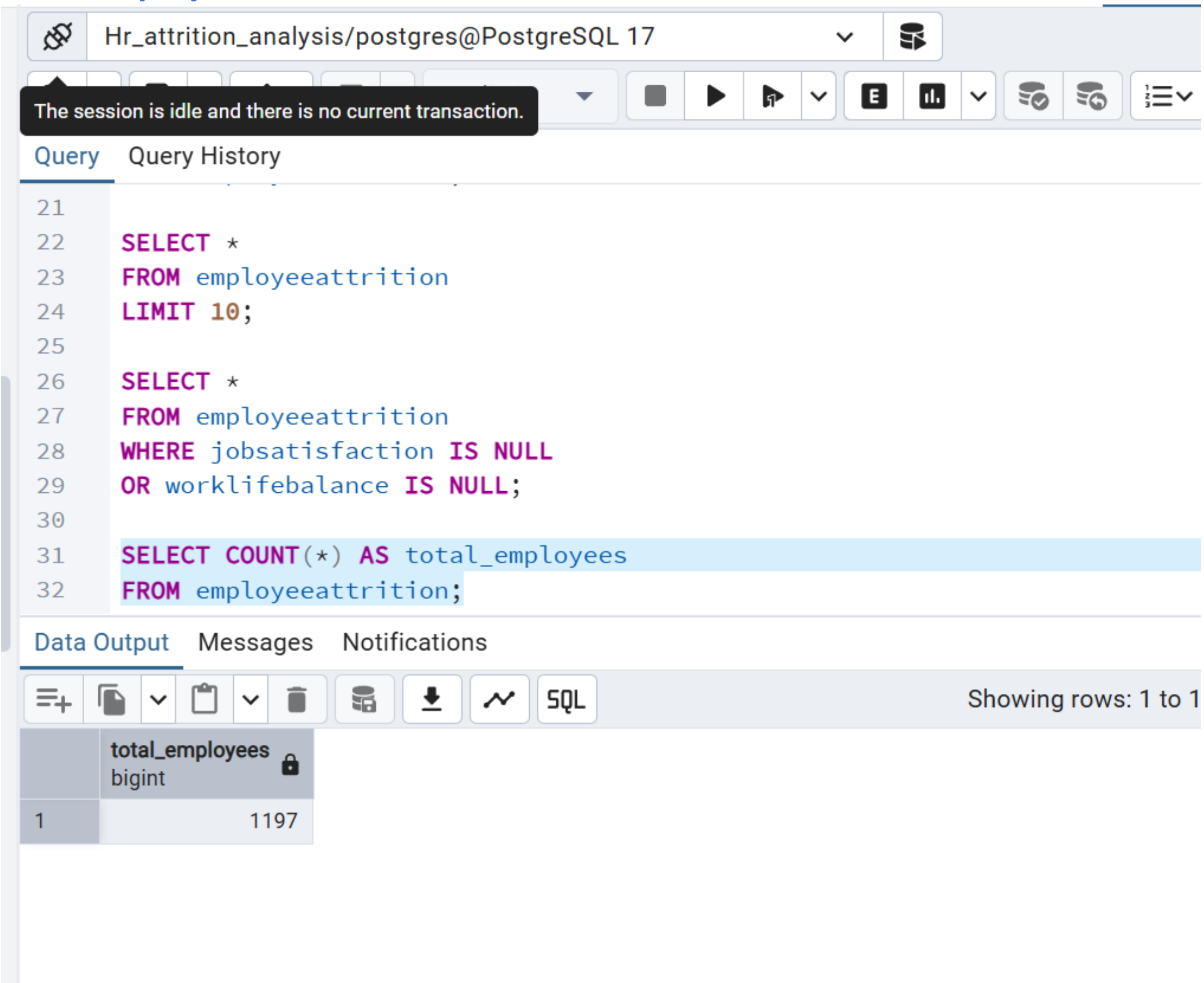


First 10 records:



	employeeid character varying (20)	employeename character varying (100)	department character varying (50)	jobrole character varying (100)	gender character varying (20)	age integer	monthlyincome numeric
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	M	38	10524
2	EMP0002	Jesse Robbins	It	Software Engineer	F	53	22226
3	EMP0003	Terri Huynh	It	Software Engineer	F	48	13935
4	EMP0004	Adam Green	Hr	Recruiter	M	50	20071
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	M	35	11982
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460
7	EMP0007	Jessica Donovan	Hr	Hr Executive	F	41	4332
8	EMP0008	Mark Smith	Operations	Operations Manager	F	52	15464
9	EMP0009	Alan Brock	Hr	Recruiter	M	46	14361
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	M	25	15108

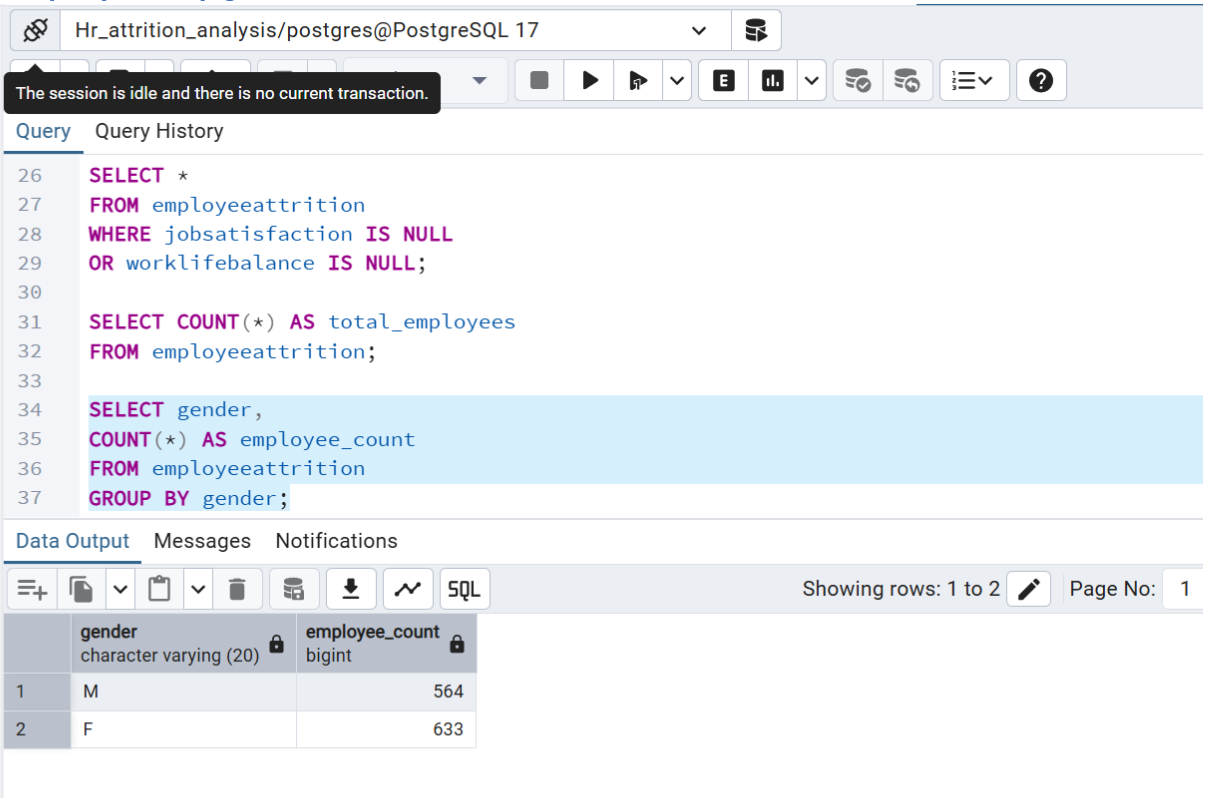
Total employees:



Description

- The total number of employees present in the organization was calculated.
- This analysis provides an overview of the workforce size available in the dataset.
- The result serves as the baseline for all further attrition and workforce analyses.

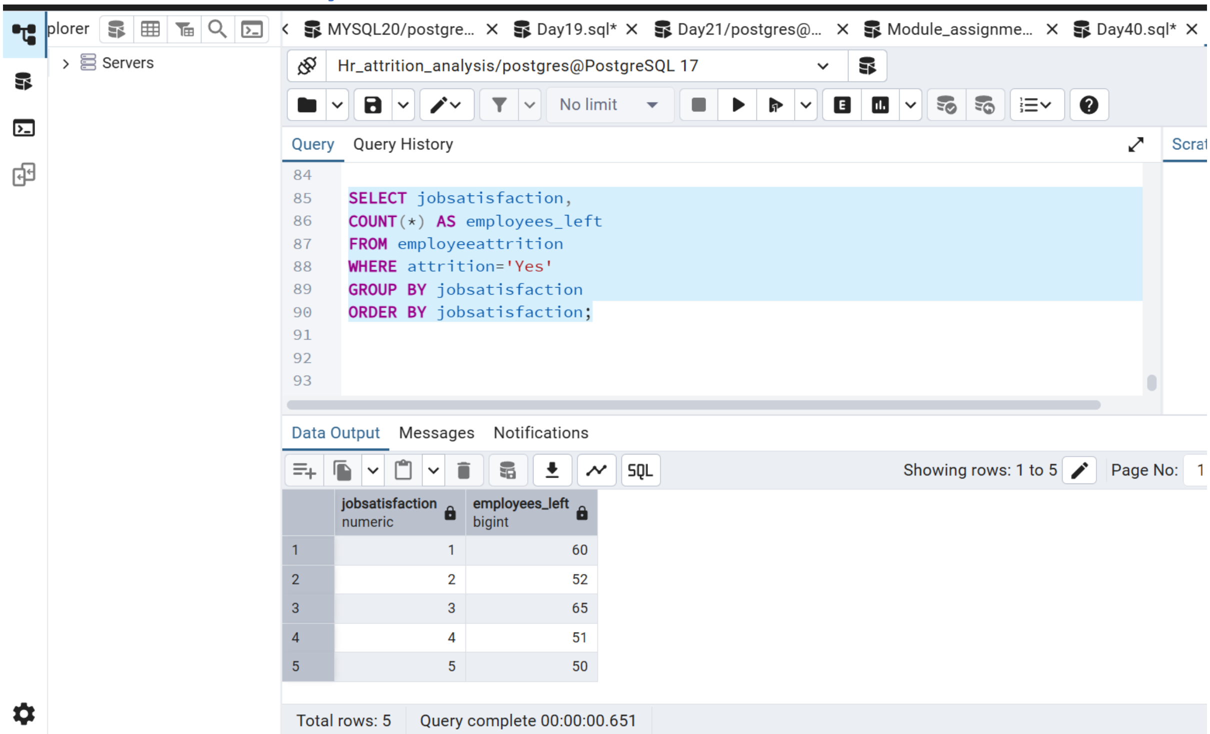
Employees by gender:



Description

- The workforce was analyzed based on gender distribution within the organization.
- This analysis helps understand the representation of different gender groups across the employee population.
- The findings provide insights into workforce diversity and support demographic analysis for organizational planning.

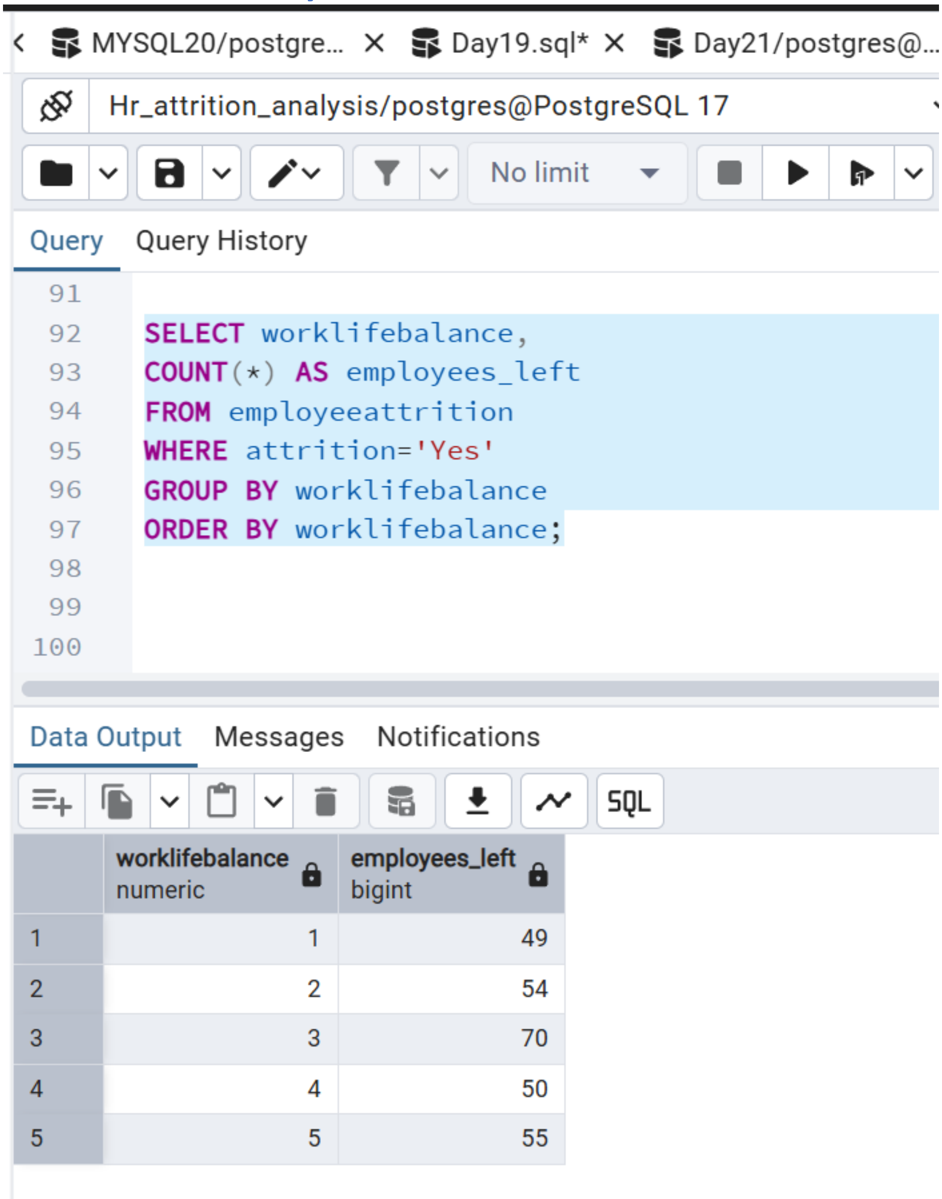
Job Satisfaction Analysis



Description

- Employee attrition was analyzed based on job satisfaction levels.
- The analysis helps identify the relationship between employee satisfaction and turnover.
- The results support initiatives aimed at improving employee engagement and retention.

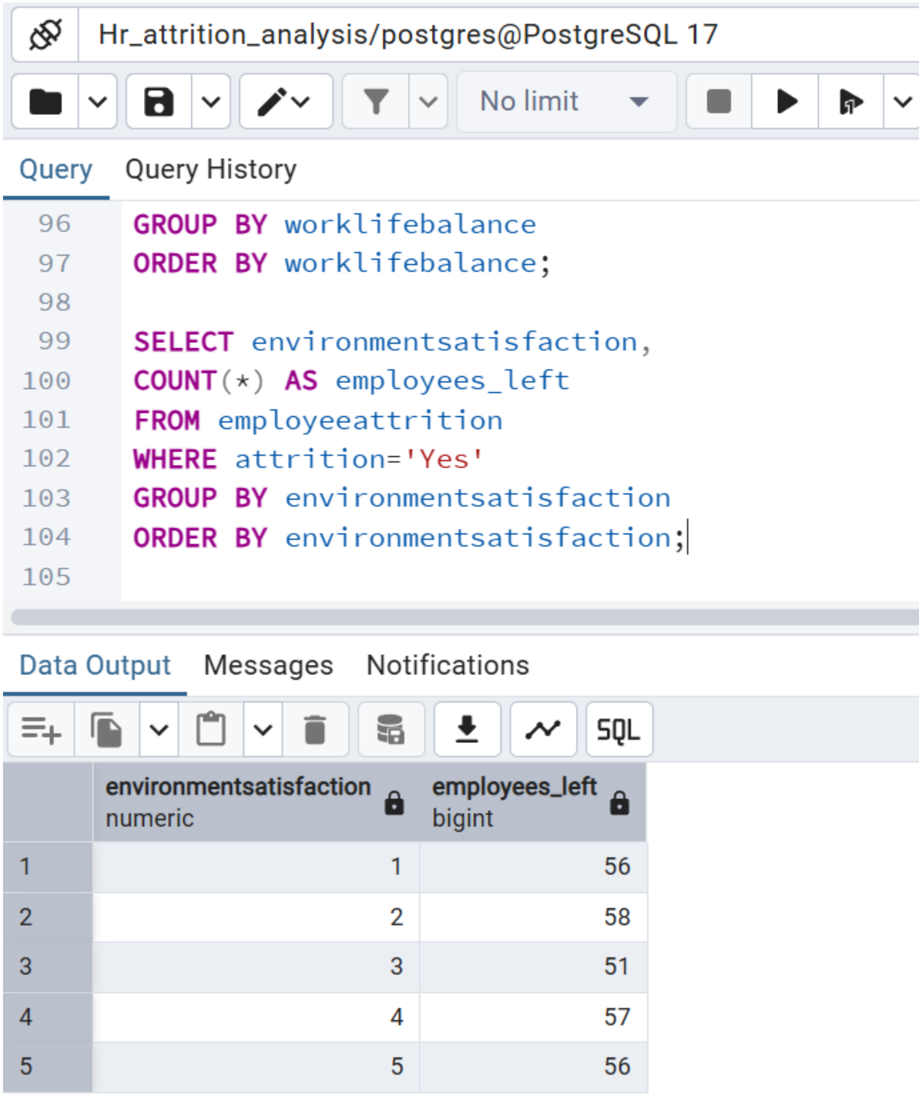
Work life balance by attrition



Description

- Employee turnover was examined across different work-life balance ratings.
- The objective was to assess whether work-life balance influences employee retention.
- The findings support organizational efforts to improve employee well-being.

Environment Satisfaction by Attrition



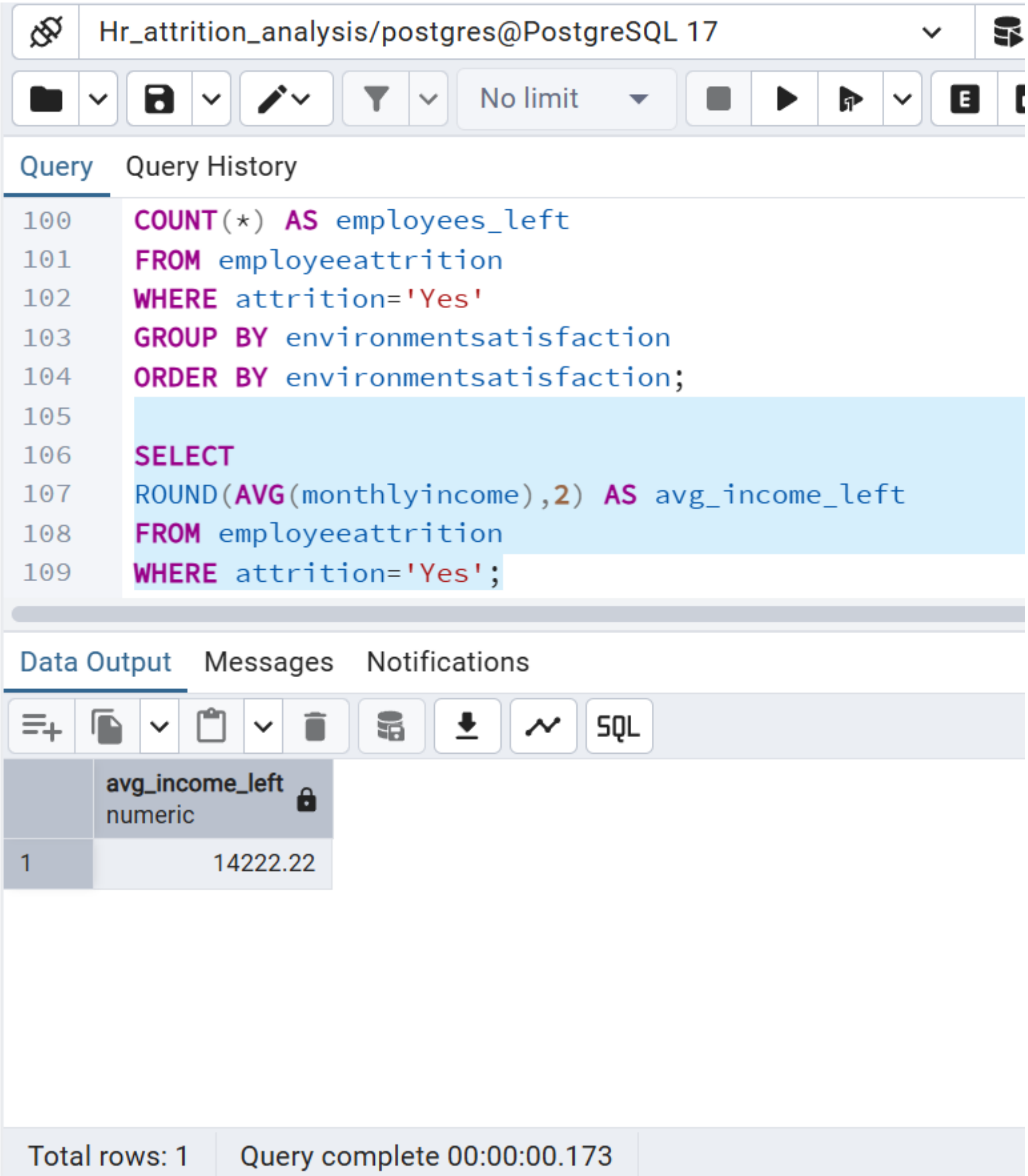

```sql
GROUP BY worklifebalance
ORDER BY worklifebalance;

SELECT environmentsatisfaction,
COUNT(*) AS employees_left
FROM employeeattrition
WHERE attrition='Yes'
GROUP BY environmentsatisfaction
ORDER BY environmentsatisfaction;
```

	environmentsatisfaction numeric	employees_left bigint
1	1	56
2	2	58
3	3	51
4	4	57
5	5	56

Description

- Employee attrition was analyzed based on workplace environment satisfaction levels.
- The analysis evaluated the impact of workplace conditions on retention.
- The results help identify opportunities to improve employee experience.

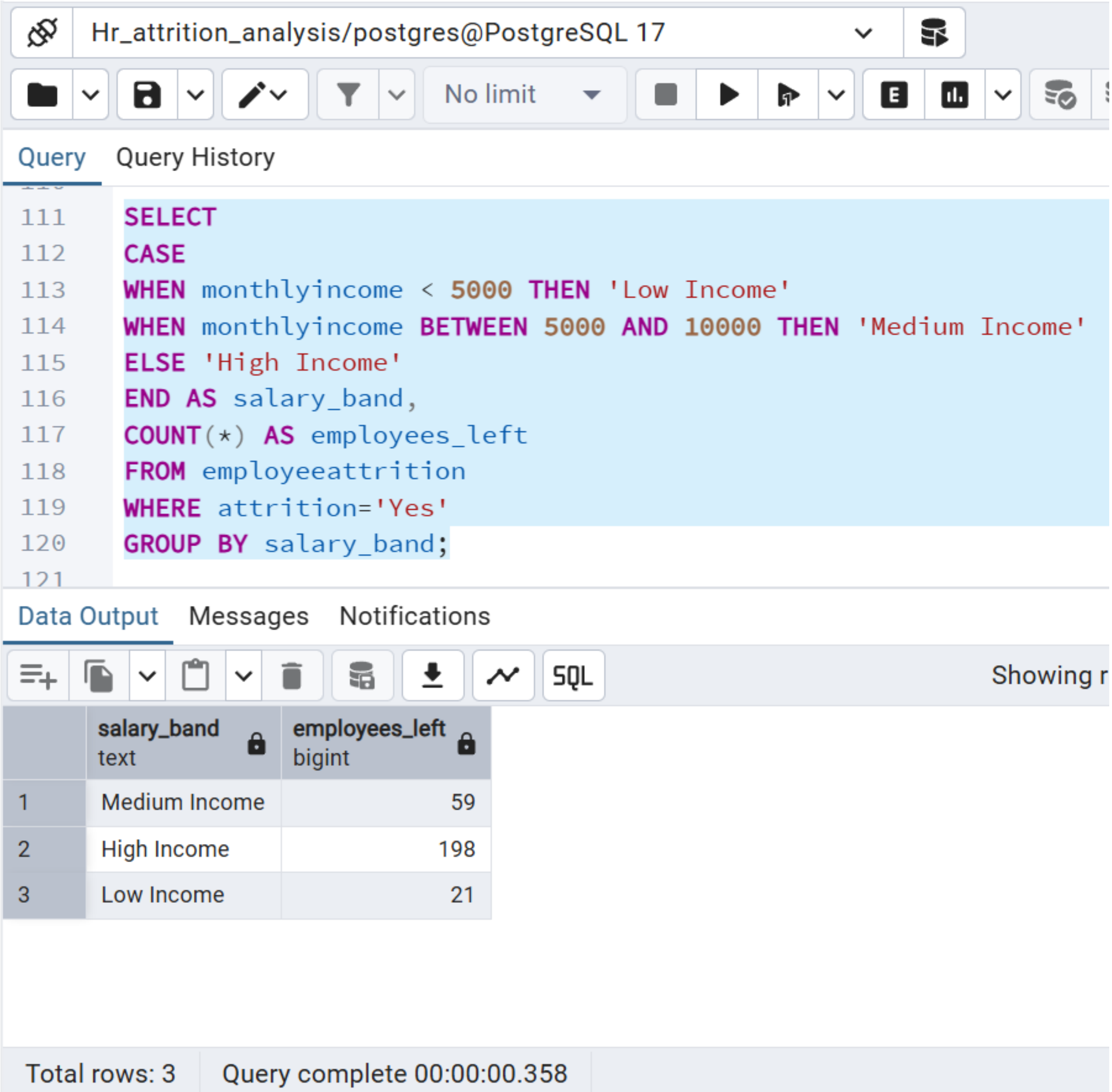
Monthly Income Analysis



Description

- The average monthly income of employees who left the organization was analyzed.
- This analysis helps assess whether compensation plays a significant role in employee attrition.
- The results support decision-making related to salary and employee retention strategies.

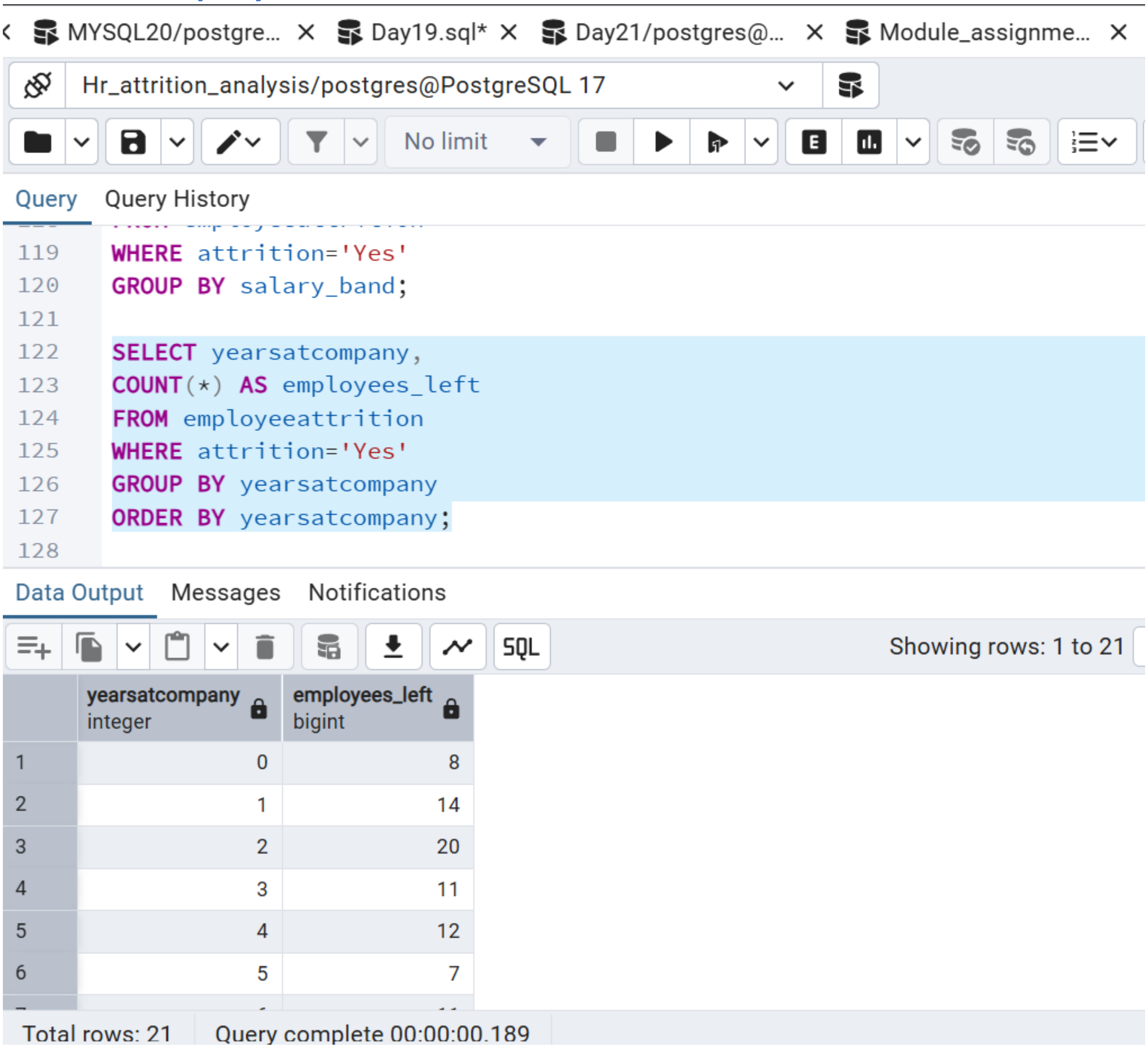
Attrition by Salary Band



Description

- Employees were categorized into salary bands to study attrition patterns.
- The objective was to identify income groups experiencing higher employee turnover.
- The findings provide insights into compensation-related retention challenges.

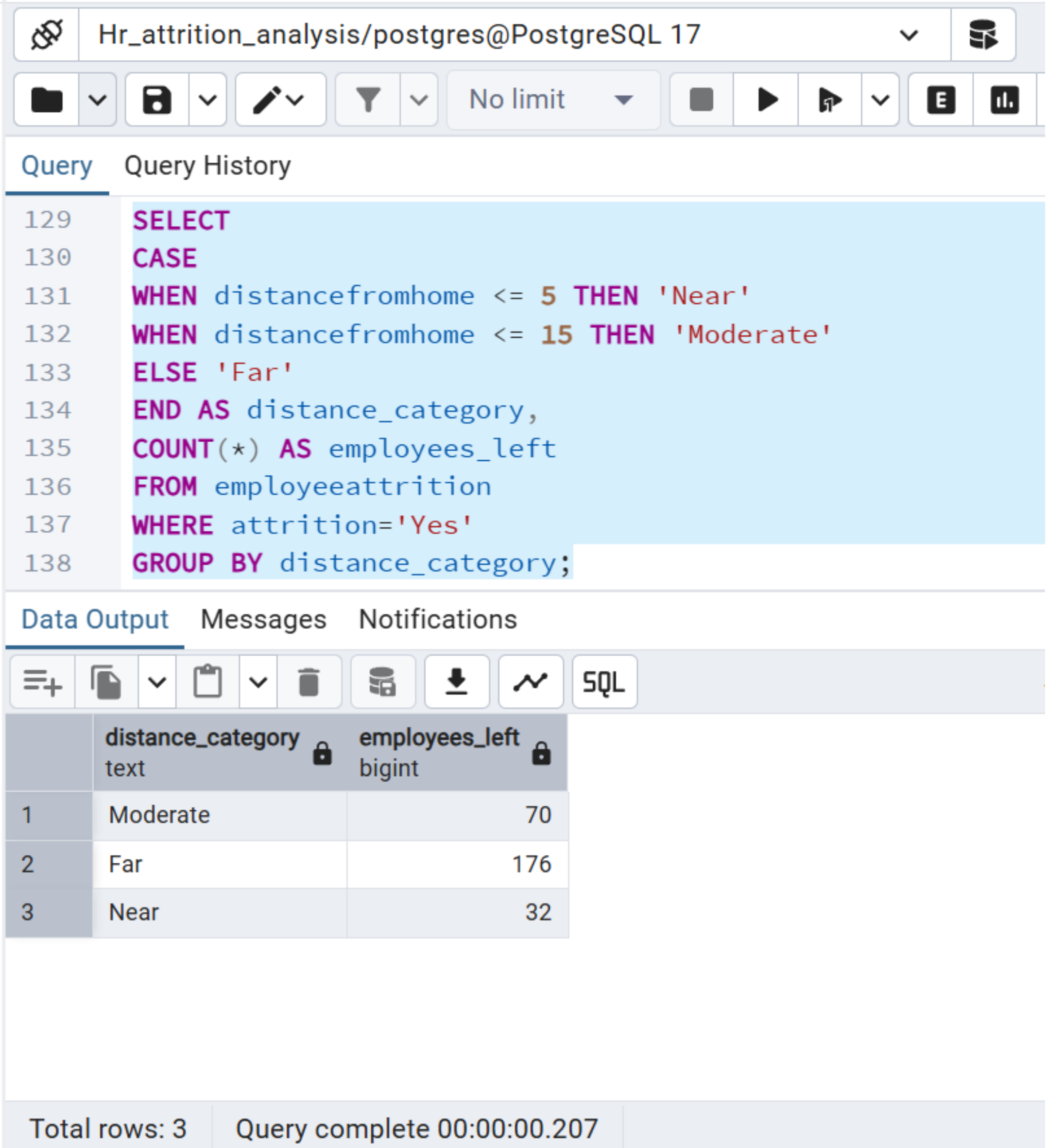
Years at Company vs Attrition



Description

- Employee attrition was analyzed based on years of service within the organization.
- The analysis helps identify at which stage employees are more likely to leave.
- Understanding tenure-based attrition supports workforce planning and retention programs.

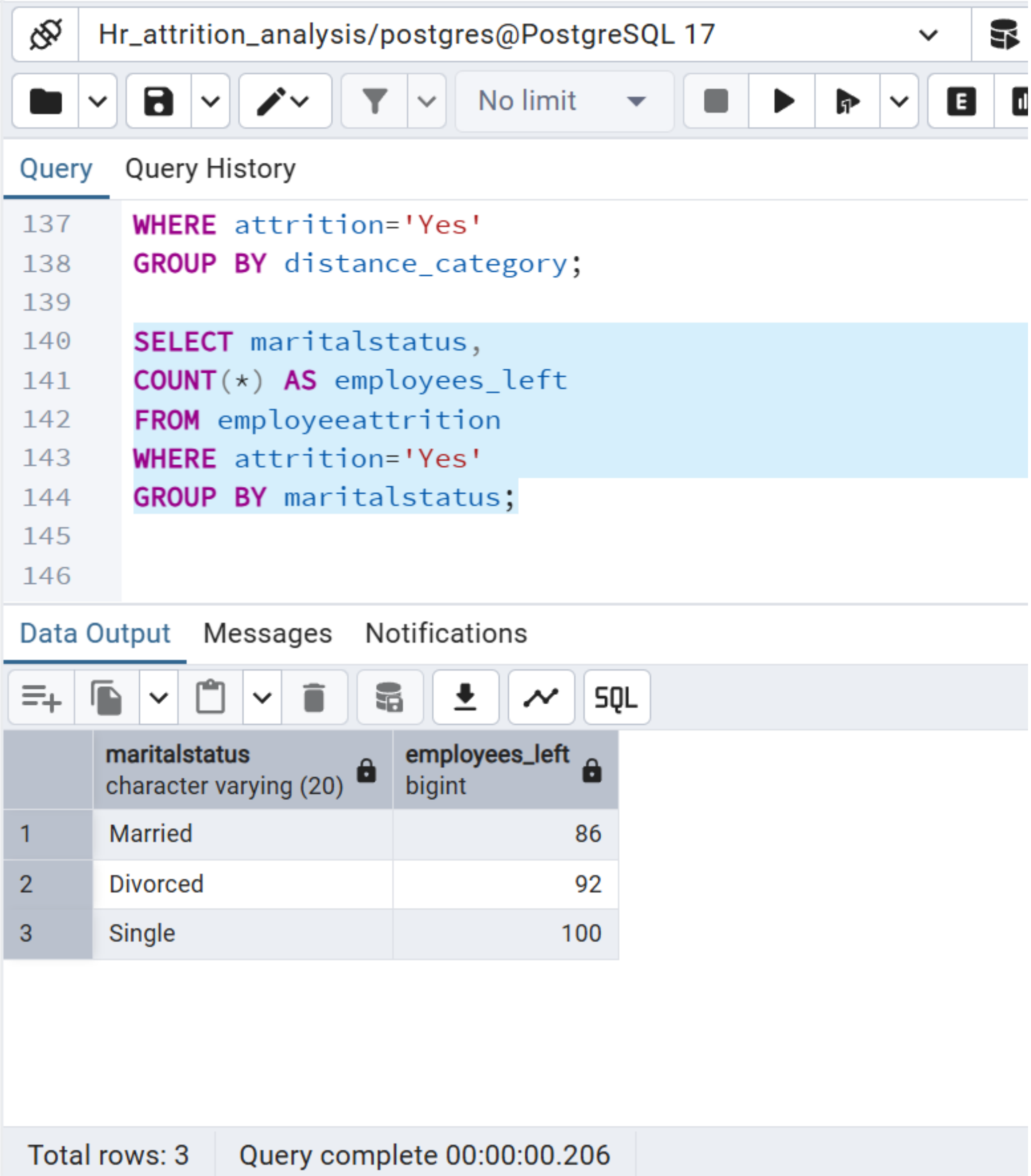
Distance from Home vs Attrition



Description

- Employee turnover was examined based on commuting distance from home to workplace.
- The objective was to determine whether travel distance influences employee retention.
- The findings help evaluate the impact of commuting challenges on workforce stability.

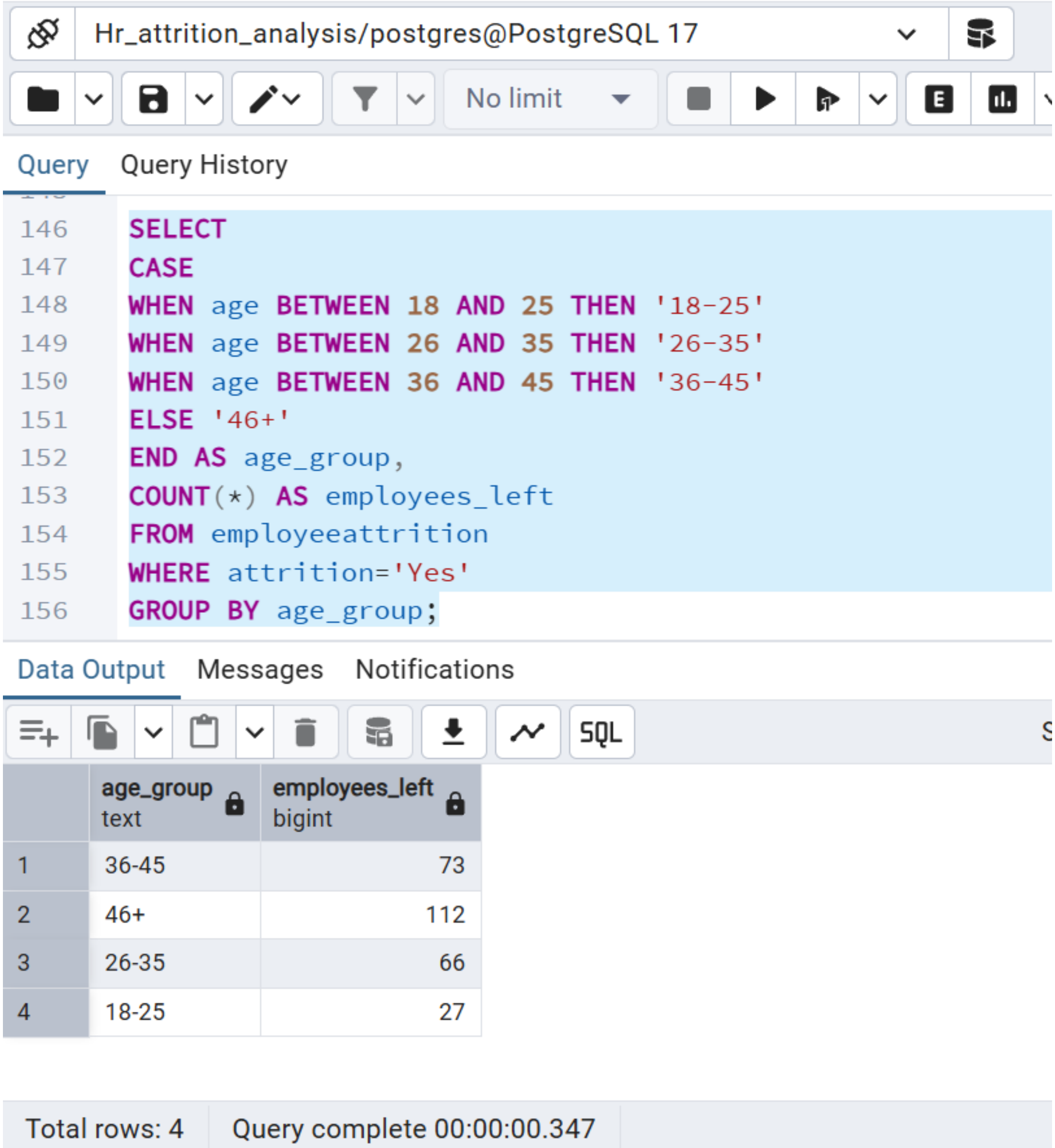
Marital Status vs Attrition



Description

- Employee attrition was analyzed across different marital status categories.
- The analysis helps understand whether personal and family-related factors affect retention.
- The results provide additional insights into employee demographics and turnover trends.

Age Group Analysis

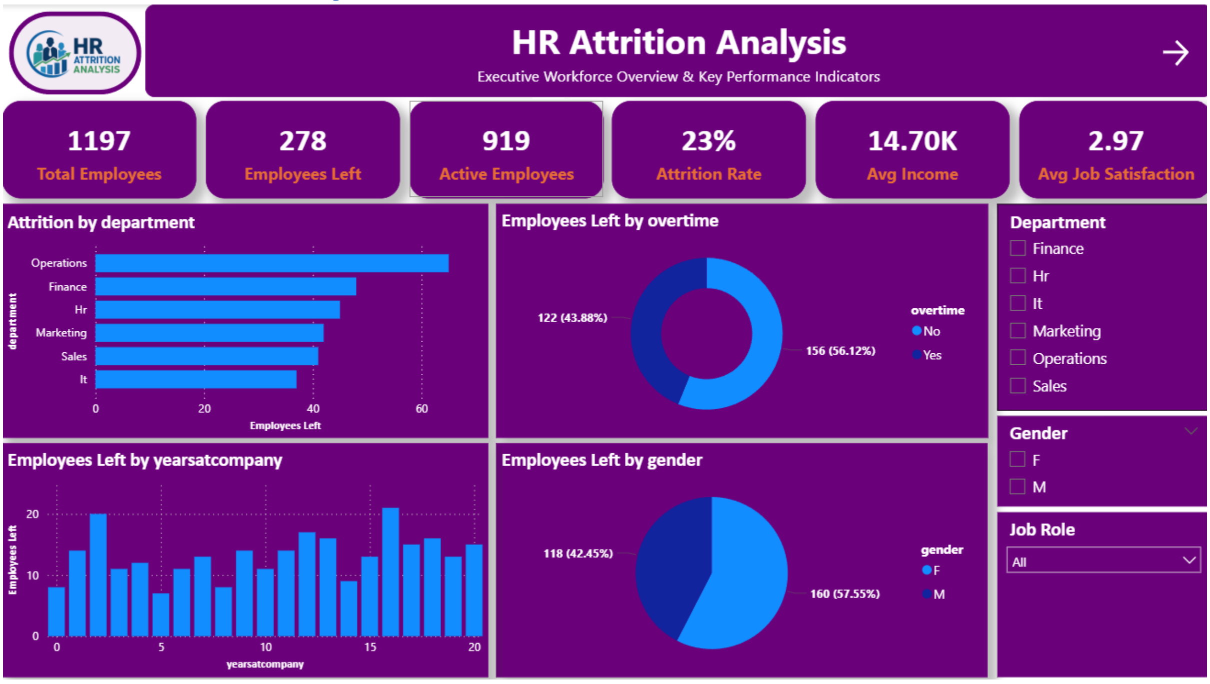


Description

- Employees were grouped into different age categories for attrition analysis.
- The objective was to identify age groups experiencing higher turnover rates.
- This analysis supports workforce planning and targeted employee engagement initiatives.

<u>**Power BI**</u>

Executive summary



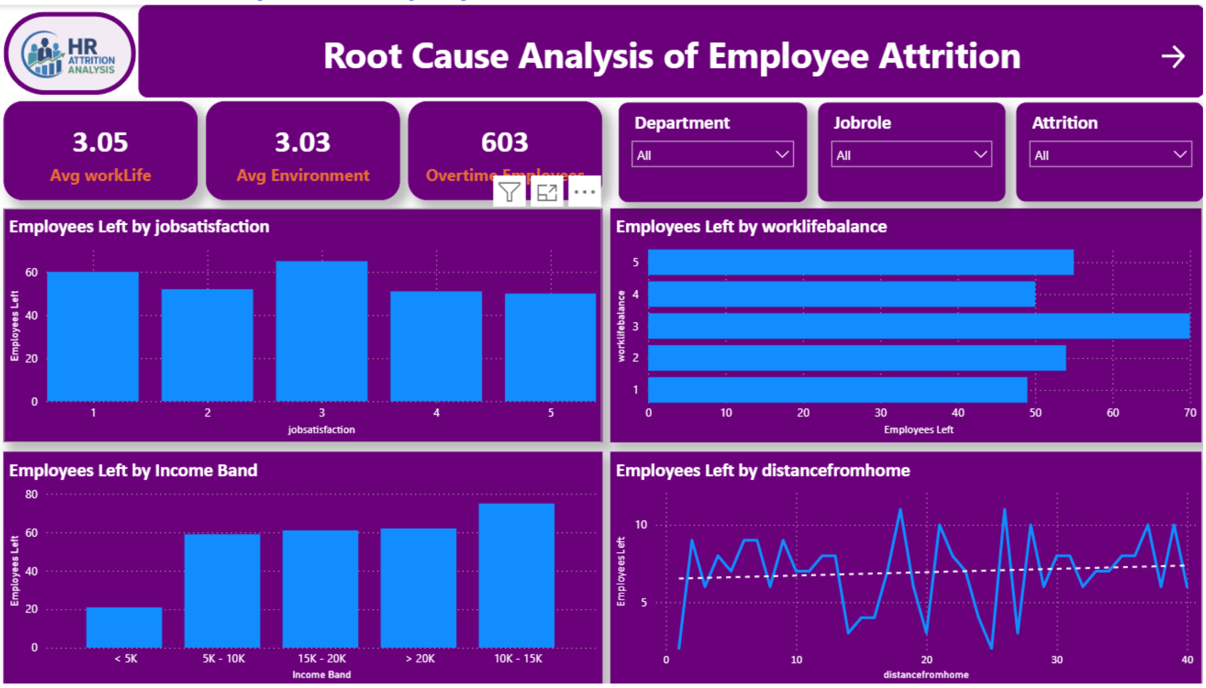
Description:

This dashboard provides an executive overview of the organization's employee attrition. It presents the key HR metrics, including Total Employees, Active Employees, Employees Left, Attrition Rate, Average Monthly Income, and Average Job Satisfaction. The dashboard also visualizes attrition across different departments, overtime status, gender, and years at the company. Interactive slicers allow users to filter the analysis by Department, Gender, and Job Role, helping HR managers quickly identify areas with higher employee turnover.

Business Outcome:

- Provides a high-level view of the organization's workforce.
- Identifies departments with the highest attrition.
- Shows the impact of overtime on employee turnover.
- Helps management monitor overall HR performance.

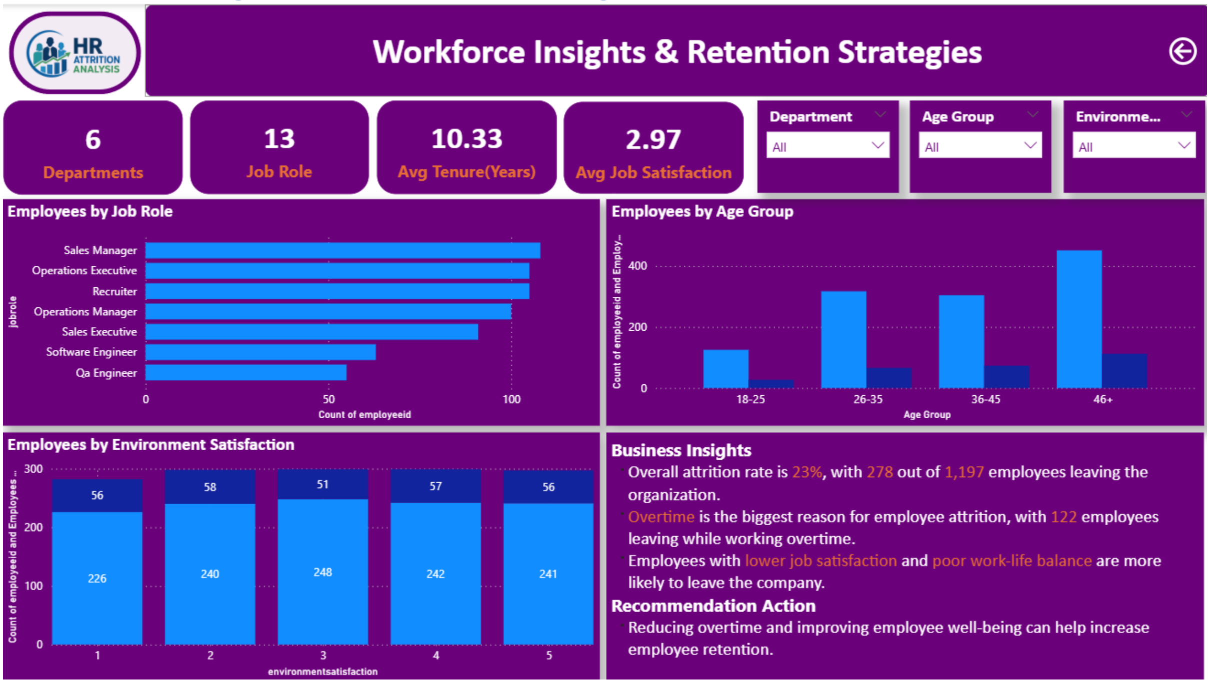
Root Cause Analysis of Employee Attrition



Description:
This dashboard focuses on identifying the major factors contributing to employee attrition. It displays KPIs such as Average Work-Life Balance, Average Environment Satisfaction, and Total Overtime Employees. Visualizations analyze employee attrition based on Job Satisfaction, Work-Life Balance, Monthly Income Bands, and Distance from Home. Slicers enable users to filter the analysis by Department, Job Role, and Attrition Status to identify patterns across different employee groups.

Business Outcome:
- Identifies the key drivers behind employee attrition.
- Highlights the relationship between overtime, job satisfaction, and work-life balance.
- Supports HR in prioritizing employee retention strategies.
- Enables department-wise analysis of attrition causes.

Workforce Insights & Retention Strategies



Description:

This dashboard provides workforce insights along with actionable recommendations for improving employee retention. It includes KPIs such as Total Departments, Total Job Roles, Average Tenure, and Average Job Satisfaction. The dashboard visualizes employee distribution by Job Role, Age Group, and Environment Satisfaction. A Key Insights section summarizes the major findings from the analysis, highlighting that overtime is the primary contributor to attrition, while lower job satisfaction and poor work-life balance further increase the likelihood of employees leaving the organization.

Business Outcome:

- Presents the overall workforce composition.
- Highlights important workforce trends across age groups and job roles.
- Summarizes the major findings from the analysis.
- Provides recommendations to reduce employee attrition through improved work-life balance and reduced overtime.

Conclusion

The HR Attrition Analysis project was developed to identify the major factors influencing employee attrition and to provide data-driven insights for improving employee retention. The analysis was performed using **Excel** for data cleaning, **PostgreSQL** for data analysis, and **Power BI** for interactive dashboard visualization.

The dashboard revealed that the organization has an **attrition rate of 23%**, with **278 out of 1,197 employees** leaving the company. The analysis identified **overtime** as the most significant factor contributing to employee attrition. Employees with **lower job satisfaction** and **poor work-life balance** were also more likely to leave the organization. Department-wise and demographic analysis further helped identify areas with higher employee turnover.

Based on these findings, the organization can improve employee retention by reducing excessive overtime, promoting a healthier work-life balance, increasing employee engagement, and regularly monitoring employee satisfaction. The interactive dashboard enables HR teams to quickly analyze attrition trends and make informed, data-driven decisions.